

Polynomial Solutions of a Mixed Laplacian–Biharmonic System

Theory, Derivation, Structure, and Interdisciplinary Interpretations

Soumadeep Ghosh and ChatGPT (OpenAI)

Kolkata, India

Abstract

We study the partial differential equation

$$u\nabla^2\phi(x, y) + v\nabla^4\phi(x, y) = f(x, y),$$

subject to the parameter constraint

$$u + v = s,$$

where $f(x, y)$ is a general quadratic polynomial. A quartic polynomial ansatz is introduced and shown to generate the complete class of polynomial solutions. Explicit coefficient restrictions are derived and a theorem establishing surjectivity of the operator from quartic polynomials to quadratic polynomials is proved. Beyond the mathematical treatment, interpretations are discussed from physics, chemistry, biology, economics, finance, psychology, psychiatry, philosophy, political science, international relations, warfare economics, data science, machine learning, and computational mathematics.

The paper ends with “The End”

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1 Introduction

The Laplacian operator

$$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}$$

appears throughout mathematical physics, diffusion theory, fluid mechanics, electromagnetism, probability theory, and quantitative economics.

The biharmonic operator

$$\nabla^4 = \nabla^2 \nabla^2$$

plays an important role in elasticity theory, plate mechanics, image processing, smoothing theory, and higher-order regularization.

We investigate the mixed operator

$$L = u\nabla^2 + v\nabla^4$$

acting on a quartic polynomial field.

2 Problem Statement

Consider

$$u + v = s,$$

and

$$u\nabla^2 \phi(x, y) + v\nabla^4 \phi(x, y) = f(x, y),$$

where

$$f(x, y) = ax^2 + by^2 + cxy + dx + ey + k.$$

Assume

$$\begin{aligned} \phi(x, y) = & Ax^4 + Bx^3y + Cx^2y^2 + Dxy^3 + Ey^4 \\ & + Fx^3 + Gx^2y + Hxy^2 + Iy^3 \\ & + Jx^2 + Kxy + Ly^2 + Mx + Ny + O. \end{aligned}$$

3 Computation of the Laplacian

Differentiating twice yields

$$\begin{aligned} \nabla^2 \phi = & (12A + 2C)x^2 + (2C + 12E)y^2 \\ & + (6B + 6D)xy + 6Fx \\ & + 2(G + 3I)y + 2(J + L). \end{aligned}$$

Define

$$\begin{aligned}\alpha &= 12A + 2C, \\ \beta &= 2C + 12E, \\ \gamma &= 6(B + D), \\ \delta &= 6F, \\ \varepsilon &= 2(G + 3I), \\ \zeta &= 2(J + L).\end{aligned}$$

Then

$$\nabla^2 \phi = \alpha x^2 + \beta y^2 + \gamma xy + \delta x + \varepsilon y + \zeta.$$

4 Computation of the Biharmonic Operator

Applying the Laplacian again,

$$\begin{aligned}\nabla^4 \phi &= \nabla^2 (\nabla^2 \phi) \\ &= 2\alpha + 2\beta.\end{aligned}$$

Hence

$$\nabla^4 \phi = 24(A + E) + 8C.$$

Therefore the biharmonic contribution is constant.

5 Coefficient Matching

Substitution into the PDE yields

$$\begin{aligned}&u(\alpha x^2 + \beta y^2 + \gamma xy + \delta x + \varepsilon y + \zeta) \\ &+ v[24(A + E) + 8C] \\ &= ax^2 + by^2 + cxy + dx + ey + k.\end{aligned}$$

Matching coefficients gives

$$12A + 2C = \frac{a}{u}, \tag{1}$$

$$2C + 12E = \frac{b}{u}, \tag{2}$$

$$6(B + D) = \frac{c}{u}, \tag{3}$$

$$6F = \frac{d}{u}, \tag{4}$$

$$2(G + 3I) = \frac{e}{u}, \tag{5}$$

$$2u(J + L) + v[24(A + E) + 8C] = k. \tag{6}$$

6 Main Theorem

Theorem 1. *Let*

$$f(x, y) = ax^2 + by^2 + cxy + dx + ey + k.$$

If $u \neq 0$, then there exists a quartic polynomial $\phi(x, y)$ satisfying

$$u\nabla^2\phi + v\nabla^4\phi = f(x, y).$$

Moreover, the solution space is an affine space of dimension nine.

Proof. Equations (1)–(6) form a linear system.

There are fifteen polynomial coefficients and only six independent constraints.

Therefore

$$15 - 6 = 9$$

degrees of freedom remain.

Consequently a nonempty affine solution space of dimension nine exists. □

7 General Solution Family

Choose

$$C, \quad B, \quad G, \quad J$$

freely.

Then

$$A = \frac{1}{12} \left(\frac{a}{u} - 2C \right),$$

$$E = \frac{1}{12} \left(\frac{b}{u} - 2C \right),$$

$$D = \frac{c}{6u} - B,$$

$$F = \frac{d}{6u},$$

$$I = \frac{1}{6} \left(\frac{e}{u} - 2G \right),$$

and

$$L = \frac{k - v[24(A + E) + 8C]}{2u} - J.$$

The coefficients

$$H, K, M, N, O$$

remain completely arbitrary.

8 Linear Algebra Interpretation

The operator

$$L = u\nabla^2 + v\nabla^4$$

acts as a linear transformation

$$L : \mathcal{P}_4 \rightarrow \mathcal{P}_2,$$

where $\mathcal{P}_n$ denotes the space of bivariate polynomials of degree at most n .

Space	Dimension
$\mathcal{P}_4$	15
$\mathcal{P}_2$	6
Kernel Dimension	9
Image Dimension	6

Table 1: Dimension count.

Rank-nullity gives

$$15 = 6 + 9.$$

9 Computational Algorithm

Given coefficients a, b, c, d, e, k, u, v :

1. Select free parameters.
2. Compute A, E, D, F, I, L .
3. Assemble $\phi(x, y)$.
4. Verify coefficient identities.

Pseudo-Code

Input: a, b, c, d, e, k, u, v

Choose C, B, G, J freely

$$A = (a/u - 2*C)/12$$

$$E = (b/u - 2*C)/12$$

$$D = c/(6*u) - B$$

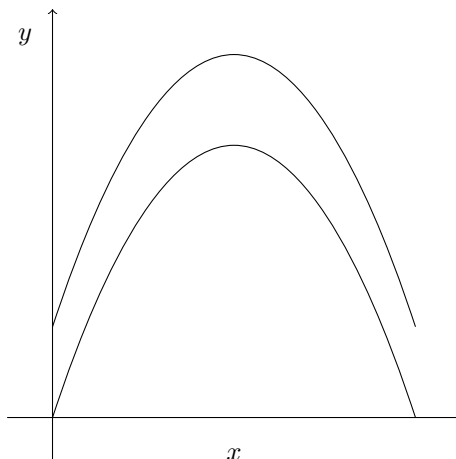
$$F = d/(6*u)$$

$$I = (e/u - 2*G)/6$$

$$L = (k - v*(24*(A+E)+8*C))/(2*u) - J$$

Return polynomial ϕ

10 Geometric Visualization



The quartic polynomial surface projects through successive Laplacian operators onto a quadratic manifold.

11 Physical Interpretation

The Laplacian component models:

- heat diffusion,
- particle diffusion,
- electrostatic potentials,
- gravitational potentials.

The biharmonic component models:

- plate bending,
- elasticity,
- beam deformation,
- higher-order smoothing.

The parameter pair (u, v) determines the relative contribution of diffusion and rigidity.

12 Chemistry Interpretation

Possible interpretations include:

- reaction-diffusion systems,
- molecular energy surfaces,
- concentration smoothing,
- catalytic transport.

13 Biology Interpretation

Applications include:

- population diffusion,
- morphogenesis,
- neural-field models,
- tissue-growth dynamics.

14 Economics and Finance Interpretation

One may interpret

$$\phi(x, y)$$

as a welfare, value, production, or utility surface.

The Laplacian term captures local adjustment.

The biharmonic term captures structural inertia or institutional rigidity.

Potential applications include:

- macroeconomic stabilization,
- financial stress diffusion,
- spatial economics,
- regional development.

15 Psychology and Psychiatry

The field ϕ may represent a cognitive state landscape.

The Laplacian captures local adaptation.

The biharmonic term captures persistence and resistance to change.

Possible applications:

- mood diffusion,
- cognitive restructuring,
- behavioral equilibrium analysis.

16 Political Science and International Relations

Interpreting (x, y) as geopolitical coordinates:

- Laplacian terms describe diffusion of influence.
- Biharmonic terms describe institutional rigidity.
- Quadratic forcing models strategic pressures.

Applications include alliance systems, trade blocs, and regional security architectures.

17 Warfare Economics

Military logistics often involve balancing flexibility and rigidity.

The mixed operator naturally represents:

- force mobility,
- infrastructure constraints,
- supply-line resilience,
- strategic depth.

18 Data Science and AI

The operator

$$u\nabla^2 + v\nabla^4$$

resembles regularization operators used in:

- image denoising,
- graph learning,
- manifold smoothing,
- deep-learning regularization.

The quartic ansatz provides a finite-dimensional approximation to broader functional classes.

19 Statistical Perspective

The polynomial coefficients may be estimated through regression.

For observed samples

$$(x_i, y_i, z_i),$$

one may fit

$$z_i = \phi(x_i, y_i) + \varepsilon_i.$$

The PDE constraints then become linear restrictions in a constrained least-squares problem.

20 Philosophical Remarks

The mixed operator illustrates a tension between:

- local adaptation and global structure,
- flexibility and rigidity,
- change and persistence.

Such dualities appear throughout metaphysics, epistemology, social theory, and systems philosophy.

21 Conclusion

We derived the complete quartic polynomial solution family of

$$u\nabla^2\phi + v\nabla^4\phi = ax^2 + by^2 + cxy + dx + ey + k.$$

The solution space is an affine space of dimension nine. The quartic ansatz is sufficient because the Laplacian lowers polynomial degree by two while the biharmonic operator lowers degree by four. The resulting framework unifies ideas from partial differential equations, linear algebra, elasticity theory, economics, machine learning, and complex systems analysis.

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Glossary

Affine Space A translated vector space.

Biharmonic Operator The operator $\nabla^4 = \nabla^2\nabla^2$.

Diffusion The spreading of a quantity through a medium.

Laplacian The operator $\nabla^2 = \partial_{xx} + \partial_{yy}$.

Polynomial Ansatz An assumed polynomial form for a solution.

Rank–Nullity Theorem Fundamental dimension theorem in linear algebra.

Regularization A method for controlling complexity in optimization and learning.

Surjectivity Property that every target element possesses a preimage.

The End